

MLP vs SonicLite-FSQ: convergence evaluation

Stopped after practical plateau | full-scale 4,096-environment experiment | dark-mode report

Executive result

At the fair iteration-7,500 comparison, MLP leads overall reward, episode survival, every reported direct tracking error, and termination robustness. SonicLite-FSQ narrows some gaps but does not overtake MLP.

26.26

MLP reward · matched final 500

19.63

Sonic reward · matched final 500

400.6

MLP episode length

388.3

Sonic episode length

Key findings

- **MLP wins the matched training-budget comparison.**
Reward is 26.26 vs 19.63; episode length is 400.6 vs 388.3.
- **Both runs reached a practical plateau, not mathematical proof of convergence.**
Final-1,000 reward slopes are under 2% of each model's mean per 1,000 iterations, but some individual metrics still drift.
- **The stopped models have different final sample budgets.**
Final conclusions use the common iteration-7,500 window. Model-specific final windows are reported separately.
- **SonicLite-FSQ is not the official SONIC architecture.**
It is a projected FSQ bottleneck ablation; it quantizes proprioception and uses different token geometry and noise settings.

Recommended model

Use the MLP checkpoint at iteration 9,000 for the strongest policy from this experiment. Preserve SonicLite-FSQ iteration 7,500 as an ablation checkpoint, not as evidence about official SONIC.

Learning dynamics

Raw scalars are faint; solid curves are trailing-100 means; dotted vertical lines mark process restarts

Mean episode reward



Mean episode length



Restart overlap handling: original streams are retained through their final recorded iterations; resumed streams are used only afterward. Raw event files remain unchanged.

Fair comparison at iteration 7,500

Trailing-500 means ending at the latest checkpoint shared by both models

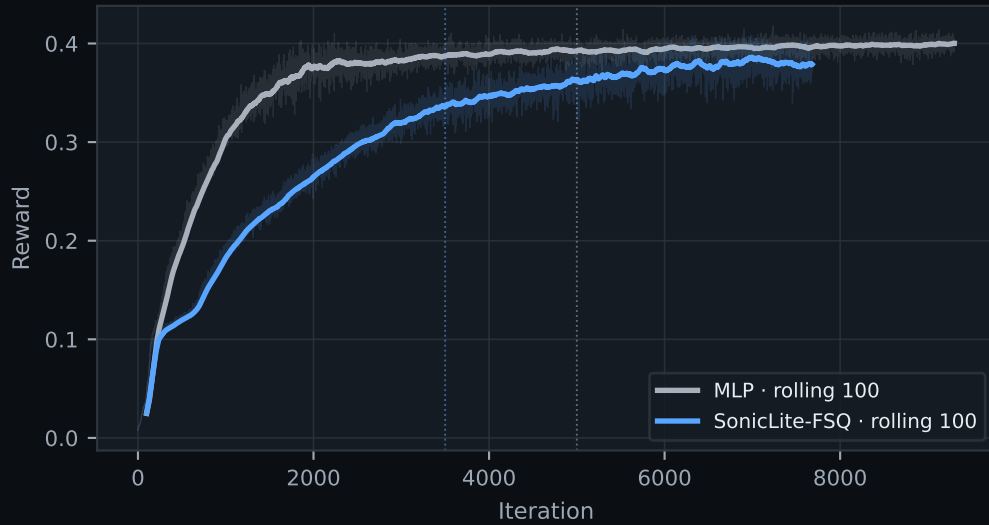
Metric	MLP	SonicLite-FSQ	Better
Mean reward	26.26	19.63	MLP
Episode length	400.6	388.3	MLP
Root orientation reward	0.3968	0.3806	MLP
Root height error	0.01238	0.01419	MLP
Root orientation error	0.000956	0.001686	MLP
Joint position error	0.10250	0.13487	MLP
Keypoint error	0.05670	0.06718	MLP
Motion completed	0.9442	0.8333	MLP
Body-position termination	0.0347	0.0911	MLP
Anchor-orientation termination	0.0114	0.0522	MLP
Policy action std	0.2495	0.3300	Descriptive
Throughput (steps/s)	33700	33398	MLP

Action std is descriptive rather than a quality score. It measures the learned Gaussian exploration distribution, not direct temporal action jitter.

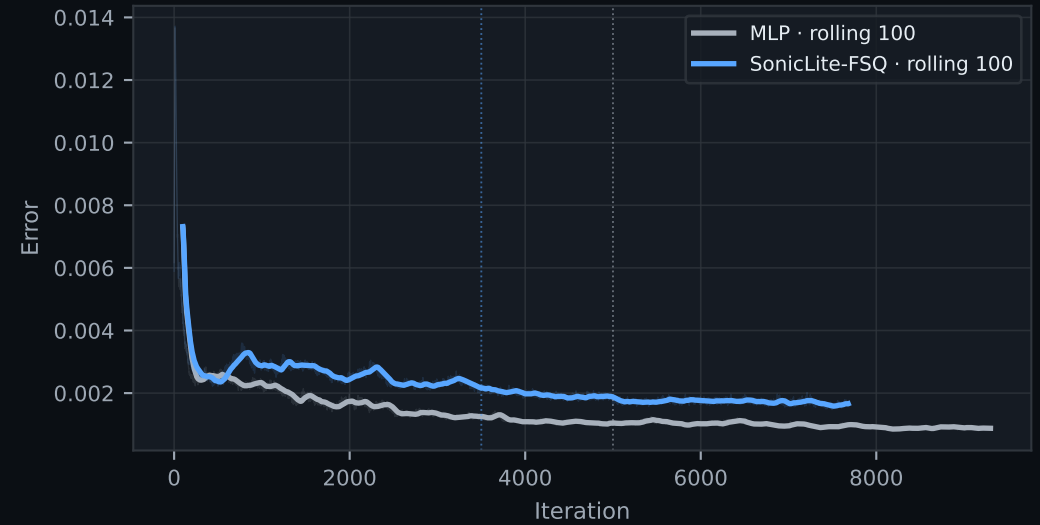
Tracking rewards and errors

Higher reward is better; lower error is better | trailing-100 curves

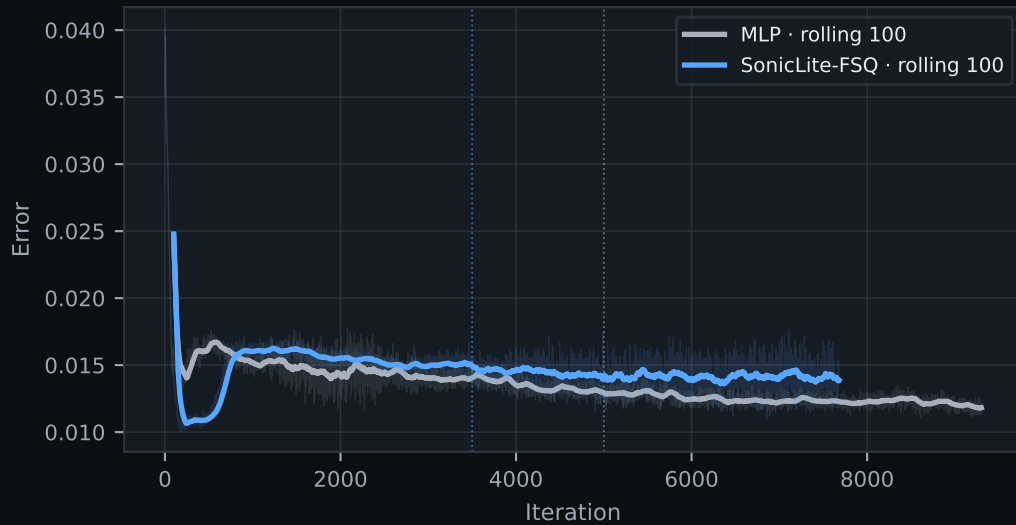
Root orientation reward



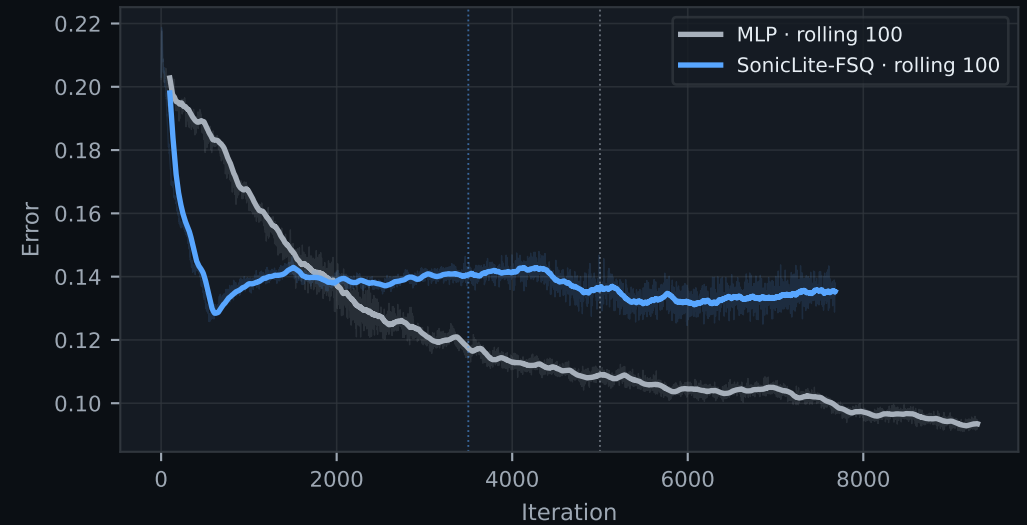
Root orientation error



Root height error



Joint position error

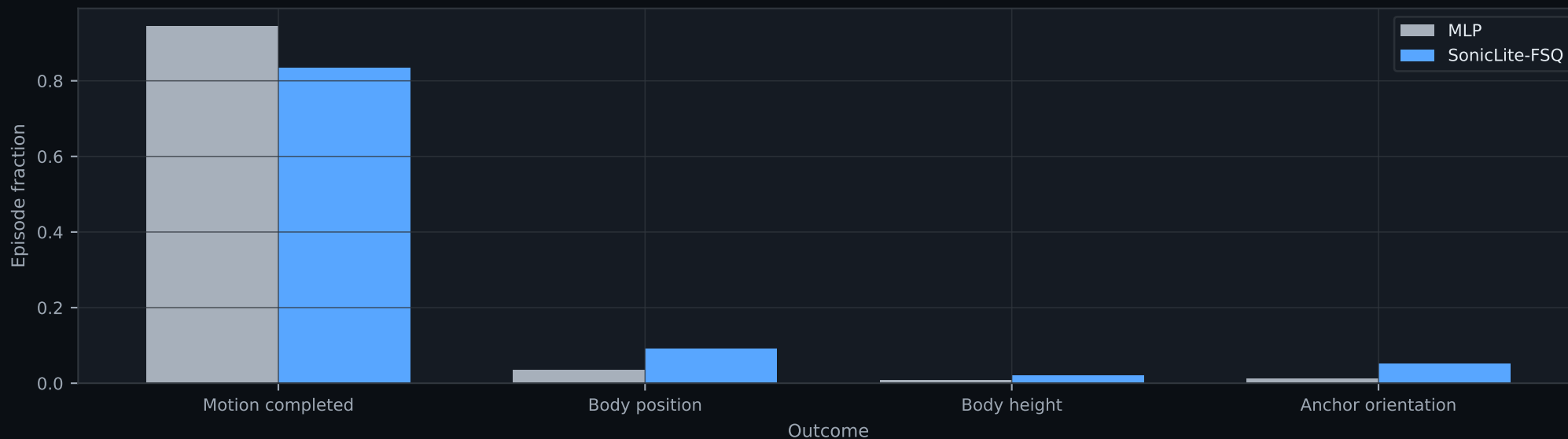


SonicLite-FSQ approached MLP on root-orientation reward but did not pass it in the matched final-500 window (0.381 vs 0.397). MLP retained lower direct errors.

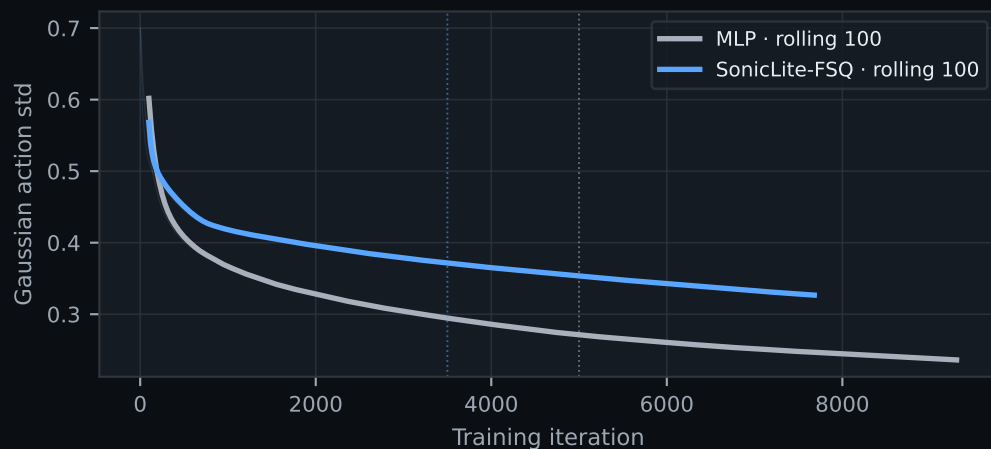
Robustness, terminations, and exploration

Matched trailing-500 window ending at iteration 7,500

Episode outcomes



Learned policy std



Training throughput



Convergence assessment

Final-1,000 linear trend | slopes are reported per 1,000 iterations

MLP

Reward: 27.17 ± 0.61 | slope $+0.49/1k$ (+1.80%)

Episode length: 402.9 ± 7.2 | slope $+0.71/1k$ (+0.18%)

Assessment: practical plateau under a $\pm 5\%$ per-1,000 heuristic; individual tracking metrics may still drift.

SonicLite-FSQ

Reward: 19.60 ± 0.90 | slope $+0.35/1k$ (+1.77%)

Episode length: 388.4 ± 13.4 | slope $-4.65/1k$ (-1.20%)

Assessment: practical plateau under a $\pm 5\%$ per-1,000 heuristic; individual tracking metrics may still drift.

Why this is not proof of full convergence

- Only one random seed was trained.
- Runs stopped at different final iterations; fair comparisons therefore use iteration 7,500.
- Sonic episode length and several error metrics were still noisy or drifting near the stop.
- No held-out motion-set evaluation was executed after stopping.

Report language

Use “practical plateau observed” rather than “proven global convergence.”

Architecture interpretation and next experiment

What this experiment does—and does not—say about SONIC

Valid conclusion

Under seed 42, 4,096 environments, rollout length 24, init action std 0.7, and the current SiliconLab reward setup, the baseline MLP outperformed the tested SonicLite-FSQ actor.

Architecture boundary

The tested actor uses a 256→40→256 projected FSQ bottleneck with eight five-dimensional codebooks and quantizes commands together with proprioception. Official SONIC uses two 32-dimensional motion tokens and gives proprioception directly to the decoder.

Noise claim boundary

This training comparison does not prove that FSQ filters observation noise. No controlled noisy-input evaluation or action-jerk test was run. Learned policy std is exploration noise, not a direct denoising metric.

Recommended next protocol

Implement a G1-only faithful SONIC path, use the advisor-specified init std 0.01 for both models, match actor parameter budgets where possible, pilot first, and evaluate clean/noisy inputs with deterministic actions.

Latest safe checkpoints

MLP: `.../mlp_convergence/2026-07-21_04-01-22_mlp_4096_resume5000_to100k_seed42/model_9000.pt`

SonicLite-FSQ: `.../sonic_fsq_convergence/2026-07-21_04-01-47_sonic_fsq_4096_resume3500_to100k_seed42/model_7500.pt`